

Def. Let $A \in M_{m \times n}(F)$ be a m by n matrix

The first elementary Row operation is: for any transposition $(i \ j) \in S_m$,

$$E_{i,j} \stackrel{\text{1st def}}{=} \begin{bmatrix} e_{(i,j)(1)} \\ e_{(i,j)(2)} \\ \vdots \\ e_{(i,j)(m)} \end{bmatrix}$$

The second elementary Row operation is: for any non-zero scalar $c \in F$, and $k \in \{1, \dots, m\}$

$$E_c \stackrel{\text{2nd def}}{=} \begin{bmatrix} e_1 \\ \vdots \\ c \cdot e_k \\ \vdots \\ e_m \end{bmatrix}$$

The third elementary Row operation is: for any scalar $c \in F$ and $1 \leq i \neq j \leq m$,

$$E_{c,i,j} \stackrel{\text{3rd def}}{=} \begin{bmatrix} e_1 \\ \vdots \\ c \cdot e_i + e_j \\ \vdots \\ e_m \end{bmatrix} \leftarrow j^{\text{th}} \text{ column.}$$

Rank of a Matrix.

Def. Given a Matrix $A \in M_{m \times n}(F)$, define

$$\text{rank } A \stackrel{\text{def}}{=} \dim \text{Im } L_A.$$

Clearly, $\text{rank } A \leq m$.

Prop. Let $A \in M_{m \times n}$, and $P \in M_{m \times m}$ and $Q \in M_{n \times n}$ be invertibles, Then

$$\text{rank } PAQ = \text{rank } A.$$

Proof. Observe that

$$\text{Im } L_{PAQ} = \text{Im}(L_P \circ L_A \circ L_Q) = (L_P \circ L_A \circ L_Q)[F^n] = (L_P \circ L_A) \underbrace{[F^n]}_{L_Q \text{ onto}} = \text{Im } L_A.$$

Moreover, since L_P is an Isomorphism, $L_P[L_A[F^n]] \cong L_A[F^n] = \text{Im } L_A$. $\square$

Thm. Given a Matrix $A \in M_{m \times n}$,

$$\text{rank } A = \dim \text{span}\{a_1, \dots, a_n\} \text{ where } a_i \text{ is the } i\text{th column of } A. \\ (\text{or } a_i = Ae_i)$$

Proof. $\text{rank } A = \text{rank } L_A = \dim \text{Im } L_A = \dim \text{span}\{L_A(e_1), \dots, L_A(e_n)\} = \dim \text{span}\{a_1, \dots, a_n\}.$

Thm. Given a Matrix $A \in M_{m \times n}$ of rank $A = r$, then there are invertibles $B \in M_{m \times m}$, $C \in M_{n \times n}$,

$$\text{such that } BAC = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

Thm. Let $T: V \rightarrow W$ and $U: W \rightarrow Z$ be linear maps on fin. dim V, W, Z . Then

$$\text{rank } U \circ T \leq \text{rank } U \text{ and } \text{rank } U \circ T \leq \text{rank } T.$$

Proof. $\text{Im } U \circ T = (U \circ T)[V] \subseteq U[W] = \text{Im } U$, so $\text{rank}(U \circ T) \leq \text{rank } U$.

$$\text{So, } \text{rank}(U \circ T) = \text{rank}(T^t \circ U^t) \leq \text{rank } T^t = \text{rank } T \quad \square$$

Def. Given $A \in M_{m \times n}(F)$, Consider the linear map

$$L_A: F^n \rightarrow F^m; X \mapsto AX$$

Then, $\ker L_A$ is the solution set of the homogeneous linear system $AX=0$.

Thm. For any kind elementary Row operation E ,

$$\ker L_A = \ker L_{EA}.$$

Proof. $AX=0$, then clearly $EAX = E0 = 0$. So $\ker L_A \subseteq \ker L_{EA}$.

Conversely, since E is invertible, $EAX=0 \Rightarrow AX=E^{-1}0=0$.

Thm. Given a m by n matrix $A \in M_{m \times n}(F)$ and $B \in F^m$,

if $S \in F^n$ satisfies $AS=B$, then

$$x \in F^n \text{ satisfies } Ax=B \Leftrightarrow x \in S + \ker L_A.$$

Proof. If $Ax=B$, then $Ax=B=AS$, so $A(x-s)=0$. Thus $x-s \in \ker L_A$, or $x \in S + \ker L_A$.

Conversely, $x \in S + \ker L_A$ be given. That is, $x = s + \gamma$ for some $\gamma \in \ker L_A$, so

$$Ax = A(s + \gamma) = B + 0 = B.$$

Corollary. $Ax=b$ has only one solution iff A is invertible.

Corollary. Given linear system $Ax=b$ where $A \in M_{m \times n}$, $x \in F^n$, $b \in F^m$,

$Ax=b$ has a solution if and only if $\text{rank } A = \text{rank } (A|b)$.

Proof. Since $Ax=b$ for some $x \in F^n$ if and only if $b \in \text{Im } A$, and

$$\text{Im } A = \text{span}\{Ae_1, \dots, Ae_n\}$$

So, proved. $\square$